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Application Credentials
Ziyi Chen
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Recommendation from Prof. Yingbin Liang	Confidential Letter of Recommendation or Evaluation	CONFIDENTIAL
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November 7, 2024

To Whom It May Concern:

This letter is to express my strongest support for Dr. Ziyi Chen's application for a faculty position in your department. Ziyi's research focuses on theoretical machine learning and large scale optimization, with a wide coverage of topics including nonconvex optimization (e.g., minimax optimization for game theory, federated learning, bilevel optimization, distributionally robust optimization and generalized smooth function optimization), trustworthy reinforcement learning (requiring safety, robustness and decentralized multi-agent decision making), and recently generalization analysis of large language model alignment. Ziyi's research has been published in top-tier machine learning conferences such as ICML, NeurIPS, ICLR as well as top journals such as Journal of Machine Learning Research (JMLR). I have collaborated with Ziyi on two research projects, and I was deeply impressed by his rapid research progress as well as deep and innovative insights into the challenging problems. Next, I will elaborate on some of Ziyi's research work that I am familiar with.

My first collaboration with Ziyi focused on the popular gradient descent ascent (GDA) algorithm for minimax optimization, which has widespread applications including game theory, adversarial training, generative adversarial network (GAN) for image/video synthesis, etc. To make GDA applicable, it is essential to ensure that the variables generated from GDA converge to a critical point. Existing works obtained such variable convergence results only for convex-type objective functions, which rules out a wide variety of popular applications such as those involving neural networks. Our work for the first time achieves variable convergence for a much larger class of objective functions with KL geometry, covering numerous widely used functions such as sub-analytic functions, logarithm and exponential functions and semi-algebraic functions. Moreover, Ziyi proposed a novel Lyapunov function with fine-tuned coefficients, and used that to derive different variable convergence rates ranging from sublinear convergence up to finite-step convergence, depending on the KL geometrical parameter. This theoretical result ensures the provable convergence guarantee for GDA to be applied to a wide variety of minimax optimization problems. This work was published at the top-tier conference of ICLR 2021.

My second collaboration with Ziyi focused on nonconvex optimization under generalized smoothness conditions. Most works on nonconvex optimization theory have been focused on the standard L-smooth objective functions, which do not capture many modern machine learning applications including distributionally robust optimization (DRO), phase retrieval and language models. Though Hessian-based generalized-smooth functions and asymmetric generalized-smooth functions have been proposed to generalize the L-smooth functions, the existing gradient-based algorithms on these generalized-smooth functions have only achieved sub-optimal complexity results for stochastic optimization. To solve this issue, Ziyi proposed a novel α -symmetric generalized-smooth function class, which by that time was the largest generalized-smooth function class that subsumes the existing L-smooth functions, Hessian-based generalized-smooth functions and asymmetric generalized-smooth functions, with applications to DRO, phase retrieval, language models, etc. To solve nonconvex optimization on the proposed large function class, he developed deterministic β -GD algorithm that achieves the optimal iteration complexity to converge to a near-stationary solution, and established the first near-optimal sample complexity result in the stochastic generalized-smooth nonconvex optimization. These convergence results are significantly more general than our initial expectation, and were validated by our experiments. This work was published at the top-tier conference of ICML 2023.

After obtaining his Ph.D. from the ECE department at the University of Utah in 2023, Ziyi joined the computer science department at the University of Maryland as a postdoctoral researcher under the mentorship of Prof. Heng Huang. In only one year, Ziyi has made an impressive contributions to theoretical reinforcement learning (RL) and minimax optimization with publications in leading venues such as ICLR 2024, ICML 2024, and NeurIPS 2024. Additionally, he also demonstrated his intellectual curiosity and versatility and ventured into reinforcement learning from human feedback (RLHF), with a new contribution on the generalization analysis of large language model alignment.

I would like to highlight Ziyi's contributions to robust RL, which is a critical framework in machine learning where an AI agent is trained in a dynamic simulation environment to ensure robust performance in different real-world settings. Despite numerous empirical successes of policy gradient algorithms for robust RL with the popular rectangular uncertainty set, there are very few theoretical works in this area. All of the existing approaches have directly optimized over the non-smooth robust value functions which yield high iteration complexity results, and further they studied only impractical settings with exact policy gradient access and small state spaces. To smooth the objective function and improve the state-of-the-art iteration complexity results, in his ICML 2024 work, Ziyi adopted an innovative approach by switching the order of the minimax optimization in the original formulation and adding entropy regularization. This yields exponential convergence of the inner policy optimization and improves iteration complexity result by the order of magnitude. Furthermore, Ziyi extended his proposed approach to stochastic setting and large state spaces that are widely used in many practical applications, and characterized the required sample complexity to achieve near-optimal global solution. These theoretical results are also supported by experiments. Later in his NeurIPS 2024 work, Ziyi defined a new and general learning framework called robust RL with general utility, which for the first time extended robust RL to general utility functions that can serve various popular goals, including the classical cumulative reward maximization, exploration, safe decision making, learning from experts' demonstrations, etc. This new framework involves a challenging nonconvex-nonconcave minimax optimization problem, for which Ziyi proposed two policy-gradient algorithms. He further derived their theoretical complexity results to achieve gradient convergence and global convergence, respectively, where the analysis is highly non-trivial. His novel contributions include the development of the gradient dominance property of general utility functions and the local invertibility property of occupancy measures.

In summary, Ziyi is an enthusiastic, well-motivated, and highly intelligent researcher with excellent mathematical skills and creativity in solving difficult problems. He has made significant contributions to both fundamental theory and engineering principles in reinforcement learning, machine learning, optimization. Further, he is keen on identifying new and interesting research topics and has remarkable passion and dedication in pursuing promising research directions. He also has a very nice personality, and is easy to collaborate with. I believe Ziyi will continue his innovation in research in his future career.

I thank you very much in advance for your consideration. Should you have any questions or require further information, please do not hesitate to contact me.

Best Regards,



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